

1 Joint, Marginal and Conditional Probabilities

In some experiments, it may be that some events are not mutually exclusive because of common elements in the sample space. These elements correspond to the simultaneous or joint occurrence of the nonexclusive events. For two events A and B , the common elements from the event $A \cap B$.

1.1 Joint Probability

The probability $P(A \cap B)$ is called the joint probability for two events A and B which intersect in the sample space. A study of a Venn diagram will readily show that

$$P(A \cap B) = P(A) + P(B) - P(A \cup B)$$

Equivalently,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \leq P(A) + P(B)$$

In other words, the probability of the union of two events never exceeds the sum of the event probabilities. The equality holds only for mutually exclusive events because $A \cap B = \emptyset$, and therefore, $P(A \cap B) = P(\emptyset) = 0$.

1.2 Conditional Probability

Given some event B with nonzero probability

$$P(B) > 0$$

we define the conditional probability of an event A , given B , by

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

The probability $P(A | B)$ simply reflects the fact that the probability of an event A may depend on a second event B . If A and B are mutually exclusive, $A \cap B = \emptyset$, and $P(A | B) = 0$.

Conditional probability is a defined quantity and cannot be proven. However, as a probability, it must satisfy the three fundamental axioms. $P(A | B)$ obviously satisfies axiom 1 by its definition because $P(A \cap B)$ and $P(B)$ are non-negative numbers. The second axiom is shown to be satisfied by letting $\Omega = A$.

$$P(\Omega | B) = \frac{P(\Omega \cap B)}{P(B)} = \frac{P(B)}{P(B)} = 1$$

The third axiom may be shown to hold by considering the union of A with an event C , where A and C are mutually exclusive. If $P(A \cup C | B) = P(A | B) + P(C | B)$ is true, then axiom 3 holds. Since $A \cap C = \emptyset$ then events $A \cap B$ and $C \cap B$ are mutually exclusive (use a Venn diagram to verify this fact) and

$$P[(A \cup C) \cap B] = P[(A \cap B) \cup (C \cap B)] = P(A \cap B) + P(C \cap B)$$

Thus, on substitution, we get

$$\begin{aligned} P[(A \cup C) | B] &= \frac{P[(A \cup C) \cap B]}{P(B)} = \frac{P(A \cap B)}{P(B)} + \frac{P(C \cap B)}{P(B)} \\ &= P(A | B) + P(C | B) \end{aligned}$$

and axiom 3 holds.

1.3 Example:

In a box there are 100 resistors having resistance and tolerance as shown in Table below. Let a resistor be selected from the box and assume each resistor has the same likelihood of being chosen. Define three events: A as “draw a 47- Ω resistor,” B as “draw a resistor with 5% tolerance,” and C as “draw a 100- Ω resistor.” From the table, the applicable probabilities are

$$\begin{aligned} P(A) &= P(47\Omega) = \frac{44}{100} \\ P(B) &= P(5\%) = \frac{62}{100} \\ P(C) &= P(100\Omega) = \frac{32}{100} \end{aligned}$$

The joint probabilities are

$$\begin{aligned} P(A \cap B) &= P(47\Omega \cap 5\%) = \frac{28}{100} \\ P(A \cap C) &= P(47\Omega \cap 100\Omega) = 0 \\ P(B \cap C) &= P(5\% \cap 100\Omega) = \frac{24}{100} \end{aligned}$$

The conditional probabilities become

$$\begin{aligned} P(A | B) &= \frac{P(A \cap B)}{P(B)} = \frac{28}{62} \\ P(A | C) &= \frac{P(A \cap C)}{P(C)} = 0 \\ P(B | C) &= \frac{P(B \cap C)}{P(C)} = \frac{24}{32} \end{aligned}$$

Tolerance

Resistance (Ω)	5%	10%	Total
22	10	14	24
47	28	16	44
100	24	8	32
Total	62	38	100

$P(A | B) = P(47\Omega | 5\%)$ is the probability of drawing a 47 Ω resistor given that the resistor drawn is 5%. $P(A | C) = P(47\Omega | 100\Omega)$ is the probability of drawing a 47 Ω resistor given that the resistor drawn is 100 Ω ; this is clearly an impossible event so the probability of it is 0. Finally, $P(B | C) = P(5\% | 100\Omega)$ is the probability of drawing a resistor of 5% tolerance given that the resistor is 100 Ω .

1.4 Total Probability

The probability $P(A)$ of any event A defined on a sample space Ω can be expressed in terms of conditional probabilities. Suppose we are given N mutually exclusive events $B_n, n = 1, 2, \dots, N$, whose union equals Ω as illustrated in Figure below. These events satisfy

$$B_m \cap B_n = \emptyset \quad m \neq n = 1, 2, \dots, N$$

$$\bigcup_{n=1}^N B_n = \Omega$$

We shall prove that

$$P(A) = \sum_{n=1}^N P(A | B_n) P(B_n)$$

which is known as the total probability of event A . Since $A \cap \Omega = A$, we may start the proof

$$A \cap \Omega = A \cap \left(\bigcup_{n=1}^N B_n \right) = \bigcup_{n=1}^N (A \cap B_n)$$

Now the events $A \cap B_n$ are mutually exclusive as seen from the Venn diagram. By applying axiom 3 to these events, we have

$$P(A) = P(A \cap \Omega) = P \left[\bigcup_{n=1}^N (A \cap B_n) \right] = \sum_{n=1}^N P(A \cap B_n) = \sum_{n=1}^N P(A | B_n) P(B_n)$$

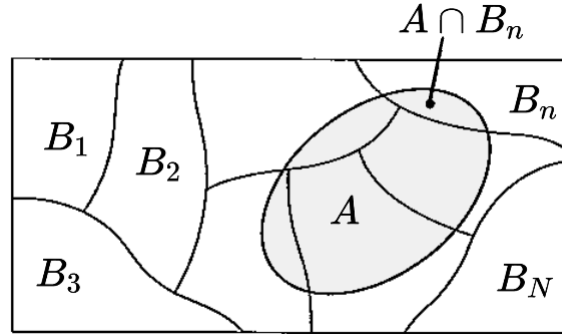


Figure 1: Venn diagram of N mutually exclusive events B_n and another event A .

1.5 Bayes' Theorem

The definition of conditional probability applies to any two events. In particular, let B_n be one of the events defined above in the subsection on total probability.

$$P(B_n | A) = \frac{P(B_n \cap A)}{P(A)}$$

if $P(A) \neq 0$, or, alternatively,

$$P(A | B_n) = \frac{P(A \cap B_n)}{P(B_n)}$$

if $P(B_n) \neq 0$. One form of Bayes' theorem is obtained by equating these two expressions:

$$P(B_n | A) = \frac{P(A | B_n) P(B_n)}{P(A)}$$

Another form derives from a substitution of $P(A)$

$$P(B_n | A) = \frac{P(A | B_n) P(B_n)}{P(A | B_1) P(B_1) + \cdots + P(A | B_N) P(B_N)}$$

for $n = 1, 2, \dots, N$.

1.6 Example:

A student takes a commuter train to get to a school's campus and to class. The probability the student will arrive at class on time is 0.95 provided the train is on time. If the train is known to be on schedule 70% of the time, what is the probability the student will be on time to class? Here we use Bayes' rule and interpret the relative frequency 70% as the probability of the train's being on time. Let C represent the event the student arrives in class and T represent the train arrives on time. The student will arrive on time if the joint event $C \cap T$ is true. The probability of this event is $P(C \cap T) = P(C | T)P(T) = 0.95(0.70) = 0.665$.

1.7 Independent Events (A Case of Two Events):

Let two events A and B have nonzero probabilities of occurrence; that is, assume $P(A) \neq 0$ and $P(B) \neq 0$. We call the events statistically independent if the probability of the occurrence of one event is not affected by the occurrence of the other event. Mathematically, this statement is equivalent to requiring

$$P(A | B) = P(A)$$

for statistically independent events. We also have

$$P(B | A) = P(B)$$

for statistically independent events. By manipulation of the above expressions, independence also means that the probability of the joint occurrence (intersection) of two events must equal the product of the two event probabilities:

$$P(A \cap B) = P(A)P(B)$$

Statistical independence is fundamental to much of our later work. When events are independent, it will often be found that probability problems are greatly simplified.

It has already been stated that the joint probability of two mutually exclusive events is 0:

$$P(A \cap B) = 0$$

If the two events have nonzero probabilities of occurrence, then, we easily establish that two events cannot be both mutually exclusive and statistically independent. Hence, for two events to be independent they must have an intersection $A \cap B \neq \emptyset$.

If a problem involves more than two events, those events satisfying either $P(A_i | A_j) = P(A_i)$ or $P(A_i \cap A_j) = P(A_i)P(A_j)$ are said to be independent by pairs.

1.8 Example

Two numbers x and y are selected at random between zero and one. Let the events A , B , and C be defined as follows:

$$A = \{x > 0.5\}, \quad B = \{y > 0.5\}, \quad \text{and } C = \{x > y\}.$$

Are the events A and B independent? Are A and C independent? Draw the sample spaces of events A , B and C to notice that

$$P[A \mid B] = \frac{P[A \cap B]}{P[B]} = \frac{1/4}{1/2} = \frac{1}{2} = P[A],$$

so events A and B are independent. Again we have that the "proportion" of outcomes in S leading to A is equal to the "proportion" in B that leads to A . Next, we have

$$P[A \mid C] = \frac{P[A \cap C]}{P[C]} = \frac{3/8}{1/2} = \frac{3}{4} \neq \frac{1}{2} = P[A],$$

so events A and C are not independent. We can see that knowledge of the fact that x is greater than y increases the probability that x is greater than 0.5.